

Computer Netzwerke und verteilte Systeme Summer 2026



Exercise 12: Time in Distributed Systems

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Deadline: The deadline for this exercise is **14.07.2026, 23:59** and in total **24 bonus points** can be achieved.

Bonus points: *Computer Netzwerke und verteilte Systeme* features a bonus point system to acknowledge first-rate performance. Exercise points count as bonus. Throughout the entire course, a bonus of up to a maximum of 0.7 can be obtained. The bonus does not help in passing the exam: You need to pass the exam with a 4.0 or better before any bonus is applied.

Group size: The recommended group size for submissions is 3. You may also submit your exercises as a group of 2 or alone if you do not find a group.

Important: Every theoretical task and subtask require proper sources. Submissions without proper sources will automatically receive 0 points.

For most of the task the slides of the lecture will be enough. For **research** task, use the materials provided in the lecture, or the internet.

Refrain from copying and pasting your answers, as this is considered plagiarism and will have **consequences**, as will using AI tools to create your answers. Instead, explain your answers in your own words and cite your sources!

Write down your answers in a legibly and unambiguously way, either by creating formatted documents (LaTeX, Word, typst, etc.) or by handwriting (**ensure that your handwriting is legible**).

Refrain from using colors that create less contrast between the text and the background. For marking tasks you may use blue and orange.

Task 1: Group Exercise: Replication in a P2P Tracker System

A small torrent community for various “Linux ISOs” runs a replicated tracker service. The primary tracker receives updates about active peers and forwards them to several backup trackers. The backup trackers are used to keep the service available if the primary or another replica fails.

- (a) What should happen if one backup tracker fails during reconfiguration?
- (b) Why must a newly added backup tracker first receive the current log/state before it becomes an active backup?
- (c) What does the ZooKeeper Atomic Broadcast (ZAB) protocol ensure in such a replicated system?
- (d) Why is the order of delivered updates important for replicated tracker state?

Task 2: Group Exercise: Lamport Clocks

- (a) Given the three processes P_0, P_1 and P_2 as shown in Table 1, we have different events taking place at different processes. With the information shown in Table 1, insert the correct local clocks for each process at every time step. In line $t = x, C_i$ describes the local clock of process P_i at time x whereas E_i names the event taking place at process P_i at time x (no event taking place if empty). For $t = 0$ you find the initial values of the local clocks. Assume that the events g, j and m include receiving messages generated by events e, i and j, respectively.

t	C_0	E_0	C_1	E_1	C_2	E_2
0	298		51		110	
1		a				
2		b				c
3		d				
4		e		f		
5						g
6				h		i
7		j				k
8		l		m		
9		n				o

Table 1: Table for Task 2 a)

- (b) This time you only have the information given in Table 2 that states the local clocks of each process at 4 time steps (+ initialization). Your task is to draw fitting Lamport Clocks. For the drawing you can refer to Figure 1 (as in the drawing you only need to state the local clocks (shown in orange) when they change). Also, label the events with arbitrary (but unambiguous) names/identifiers.

Hint: Think of when events have to happen at which process and what messages are sent.

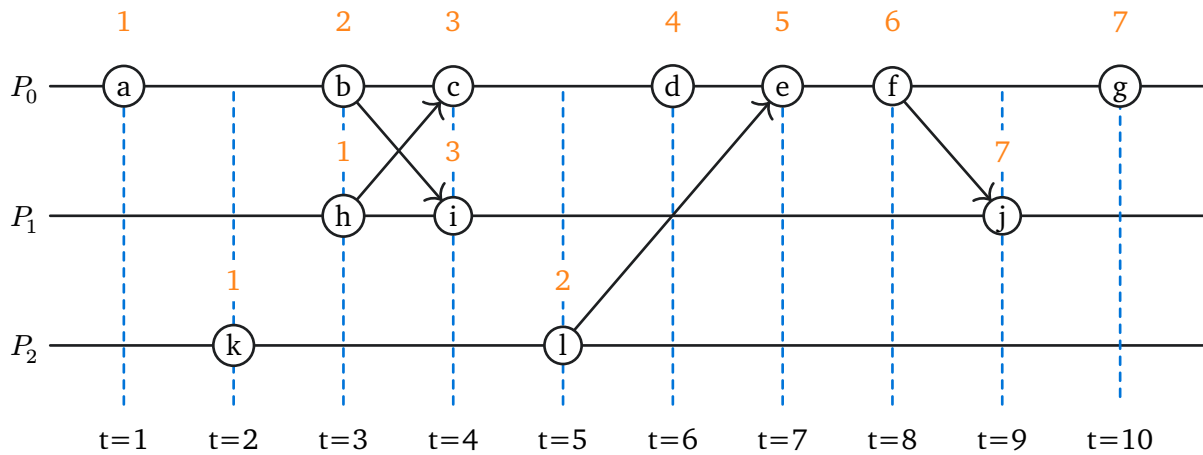


Figure 1: Lamport Clock drawing example

t	C_0	C_1	C_2
0	3	6	2
1	4	7	3
2	5	7	5
3	8	8	5
4	8	8	6
5	8	8	9

Table 2: Table for Task 2 b)

Task 3: Peer-to-Peer (5 Points)

- (a) Explain in your own words what a Peer-to-Peer system is. How does it differ from a traditional client/server system? (1.5 P)
- (b) Explain Gnutella and how query flooding works in Gnutella. (2 P)
- (c) Research: What is a “churn” problem in peer-to-peer network and why does it cause problems? (1.5 P)

Attention: Submissions without references will not be graded! Please add sources.

Task 4: Cristian's Algorithm & NTP (6 Points)

- (a) In Cristian's Algorithm, are the absolute clock values on the client and the server directly comparable? If not, what can be compared instead? (1 P)
- (b) A peer P in a P2P client wants to synchronize its clock with a trusted tracker server S . The peer sends a request at $T_{C_1} = 4200$. At $T_{C_2} = 4920$, it receives the response with the server timestamps $T_{S_1} = 4500$ and $T_{S_2} = 4700$. What value T_C does the client now set its clock to? Justify your answer with formulas and calculations. (3 P)
- (c) Practical research: On a Linux/Unix server, which service or tool is commonly used to keep the system clock synchronized? Name one example, cite one source, and explain in 1-2 sentences why this matters for servers. Bonus hint: think about logs, scheduled jobs, TLS certificates, DNSSEC, or a Pi-hole after a power outage. (2 P)

Attention: Submissions without references will not be graded! Please add sources.

Task 5: Lamport Clocks (13 Points)

- (a) Given the three processes P_0 , P_1 and P_2 as shown in Table 3, we have different events taking place at different processes. With the information shown in Table 3, insert the correct local clocks for each process at every time step. In line $t = x$, C_i describes the local clock of process P_i at time x whereas E_i names the event taking place at process P_i at time x (no event taking place if empty). For $t = 0$ you find the initial values of the local clocks. Assume that the events n, k, i, p and m include receiving messages generated by events f, g, e, j, and l respectively. (8 P)

t	C_0	E_0	C_1	E_1	C_2	E_2	C_3	E_3
0	240		151		150		200	
1		a		b				c
2		d						e
3		f		g				
4		h		i				j
5						k		l
6		m		n				
7				o				
8						p		

Table 3: Table for task 5 a)

- (b) This time you only have the information given in Table 4 that states the local clocks of each process at 5 time steps (+ initialization). Your task is to draw fitting Lamport Clocks. For the drawing you can refer to Figure 1 (as in the drawing you only need to state the local clocks (shown in orange) when they change). Also, label the events with arbitrary (but unambiguous) names/identifiers. (5 P)

Hint: Think of when events have to happen at which process and what messages are sent.

t	C_0	C_1	C_2
0	9	2	4
1	10	3	5
2	11	6	6
3	11	6	12
4	12	11	13
5	12	11	14
6	15	11	14

Table 4: Table for Task 5 b)